

BFS Traversal

Step 1: Start at A (Seeking target node K)

Legend

Complete

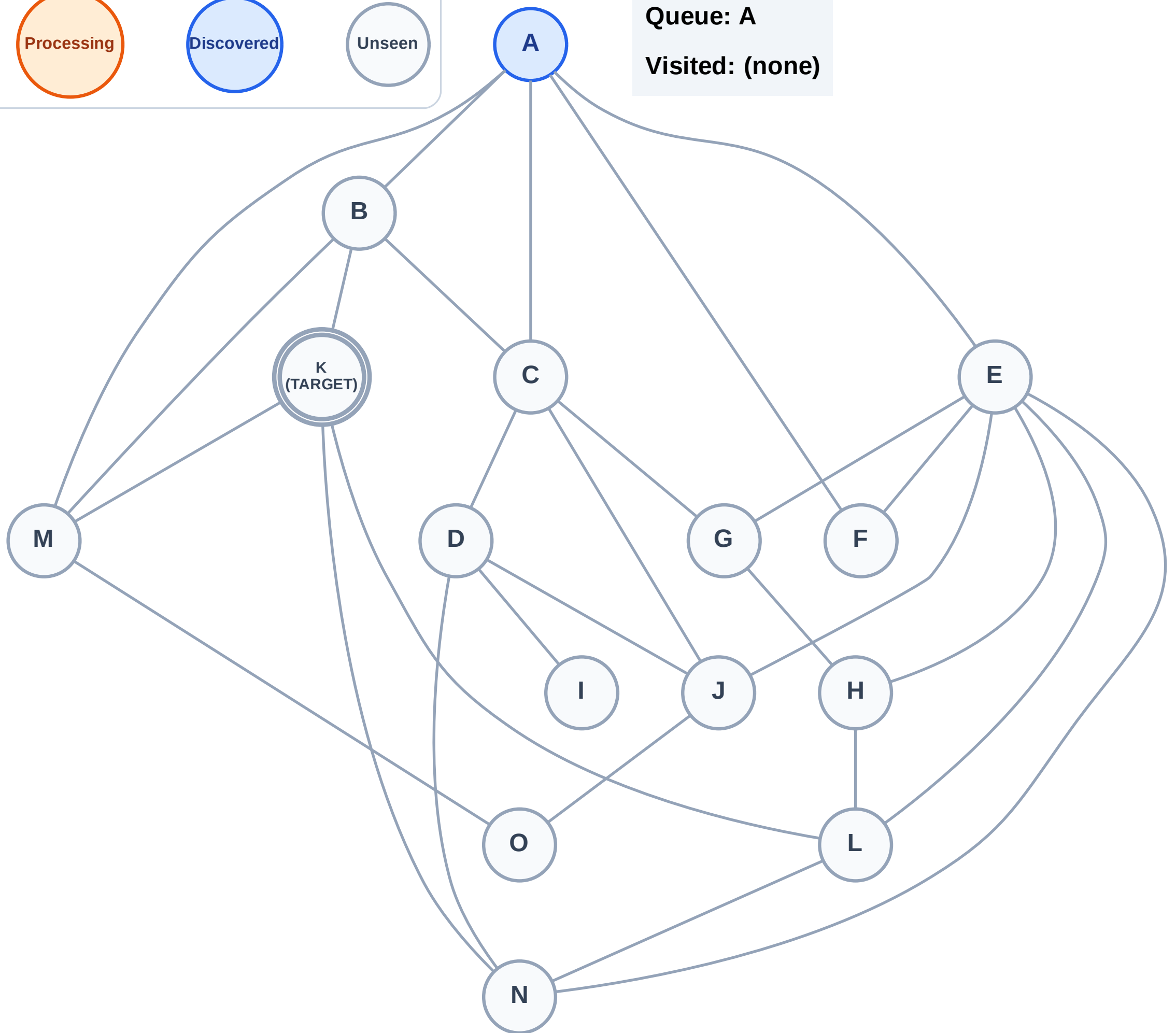
Processing

Discovered

Unseen

Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

Complete

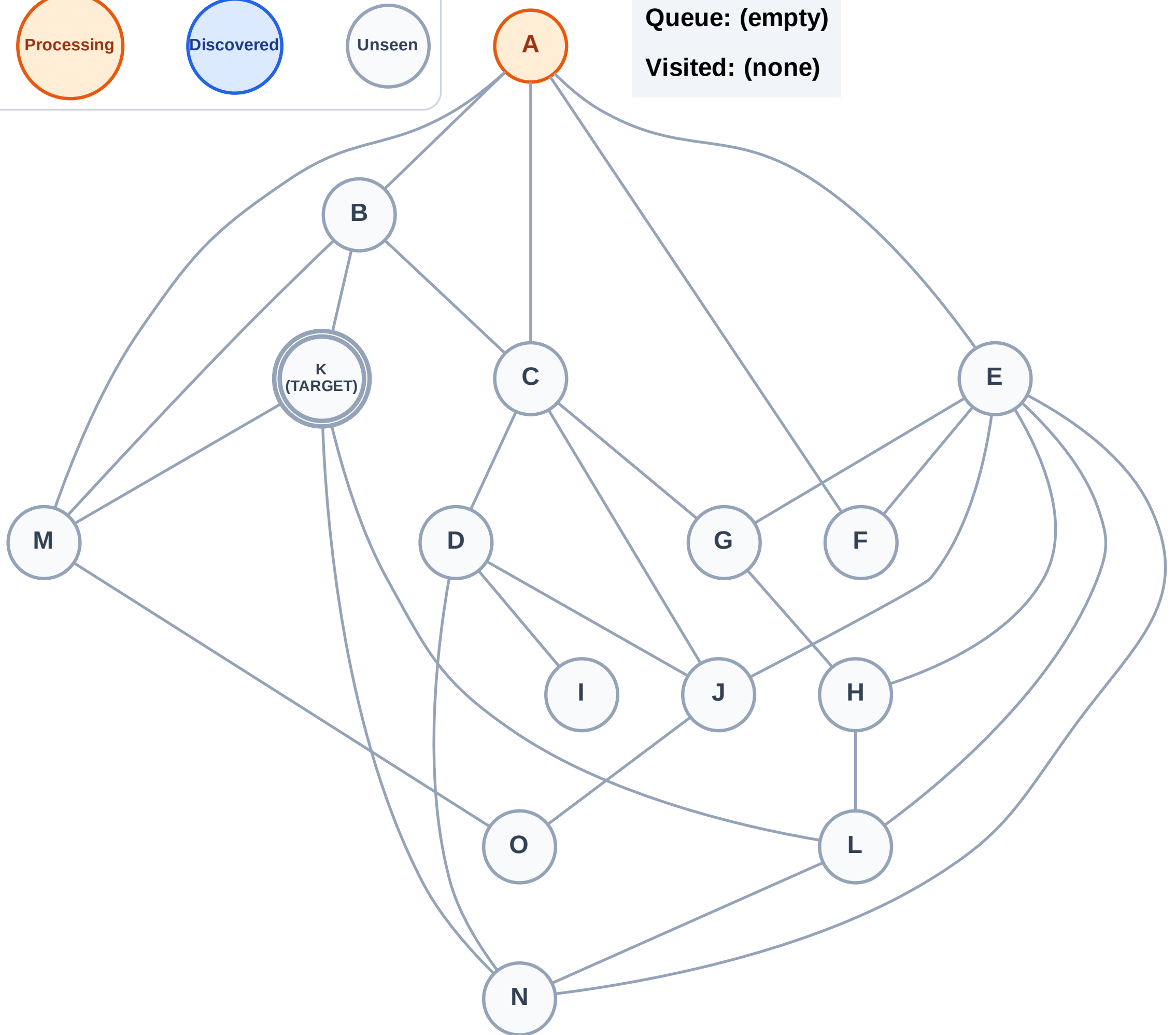
Processing

Discovered

Unseen

Queue: (empty)

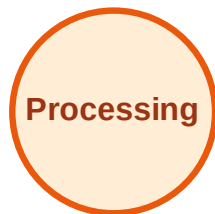
Visited: (none)



BFS Traversal

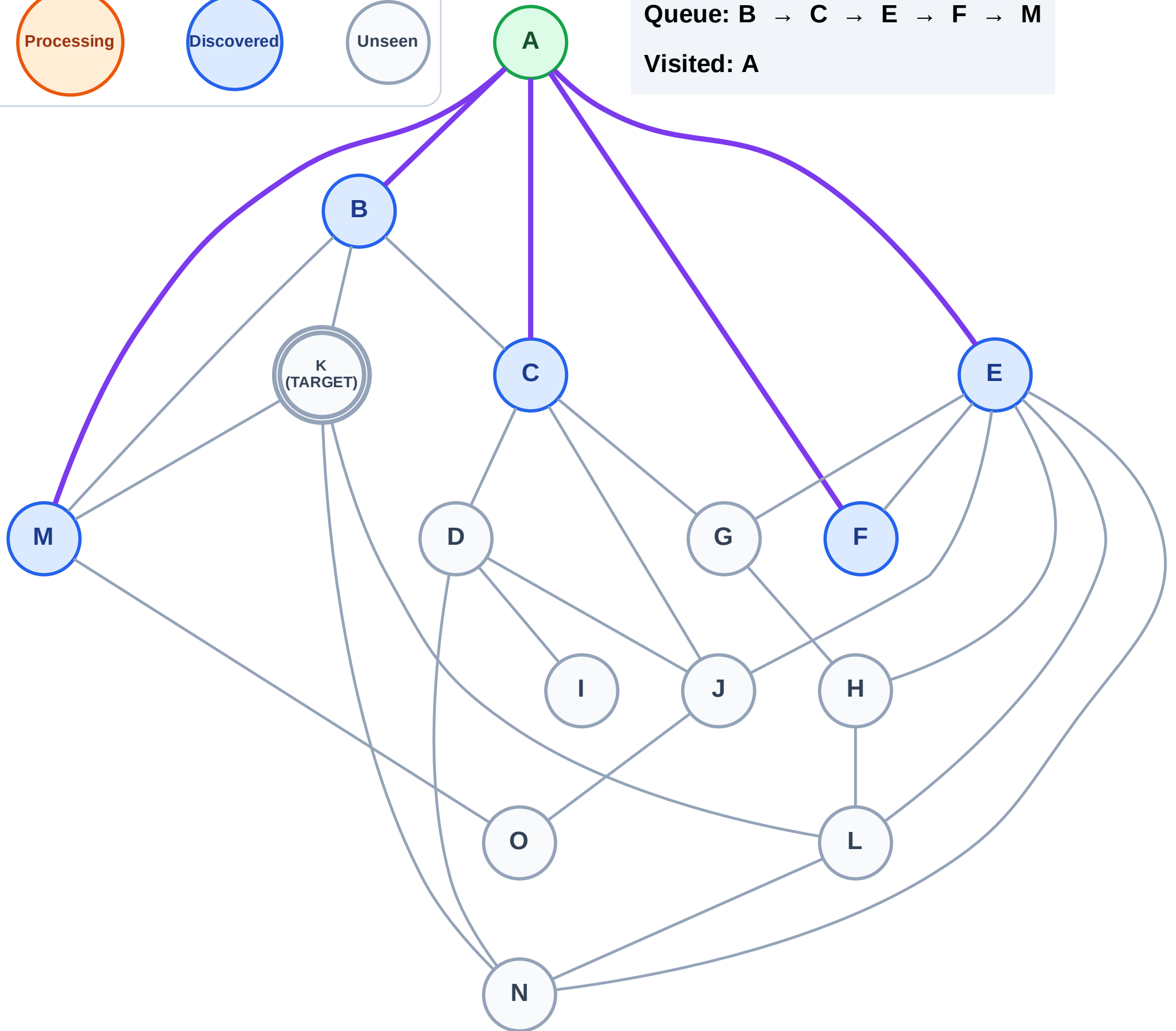
Step 3: Discover B, C, E, F, M from A

Legend



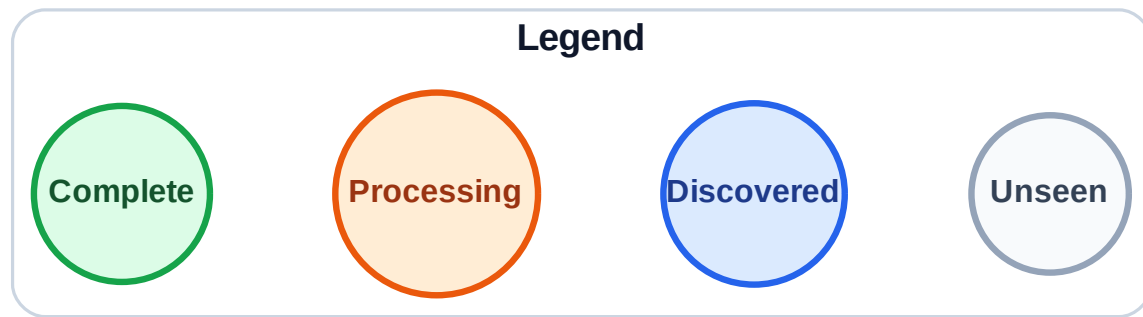
Queue: B → C → E → F → M

Visited: A

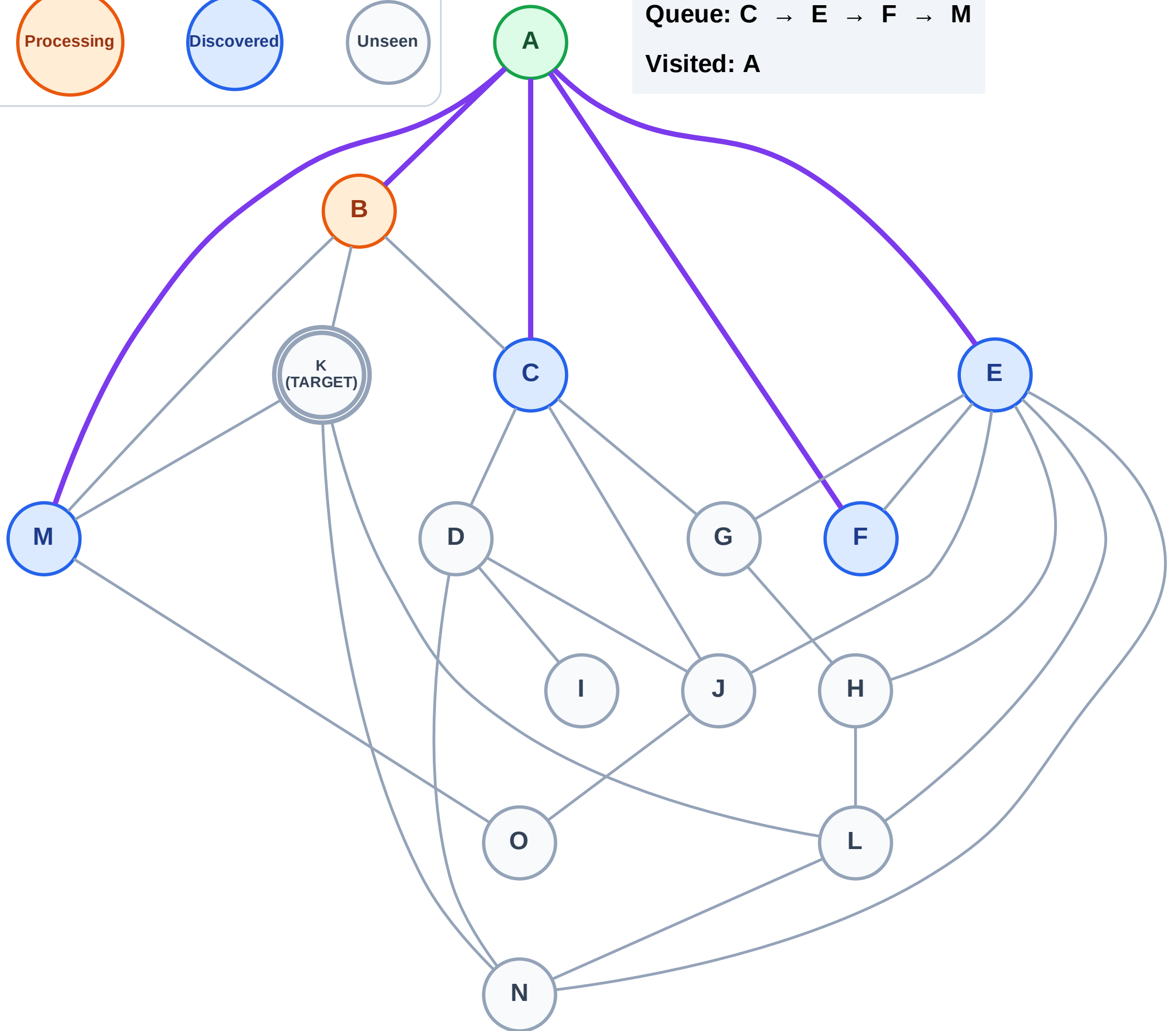


BFS Traversal

Step 4: Process node B

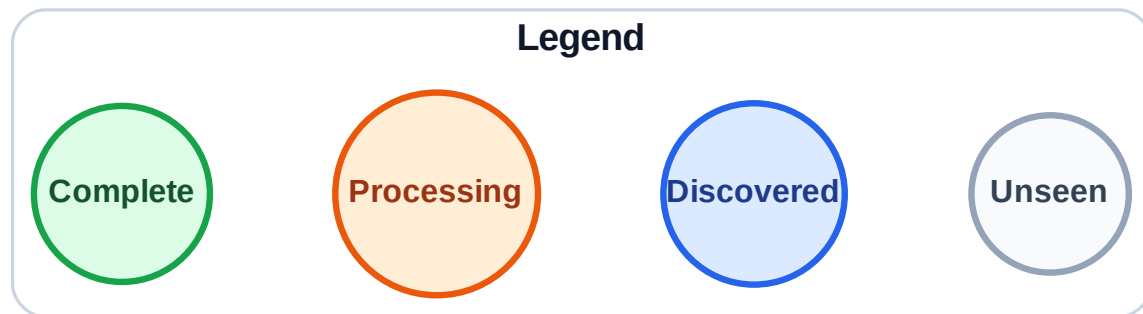


Queue: C → E → F → M
Visited: A

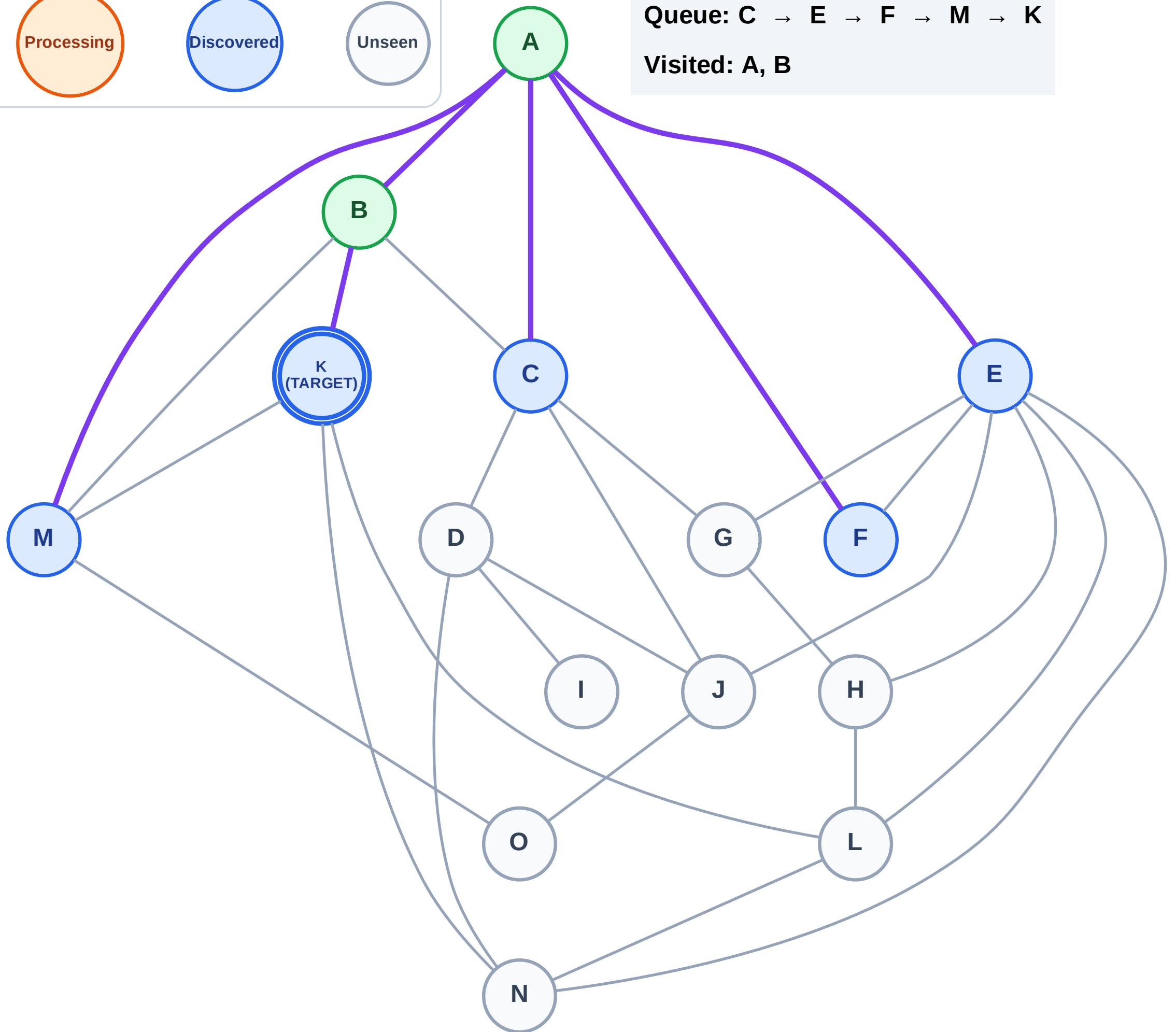


BFS Traversal

Step 5: Discover K from B

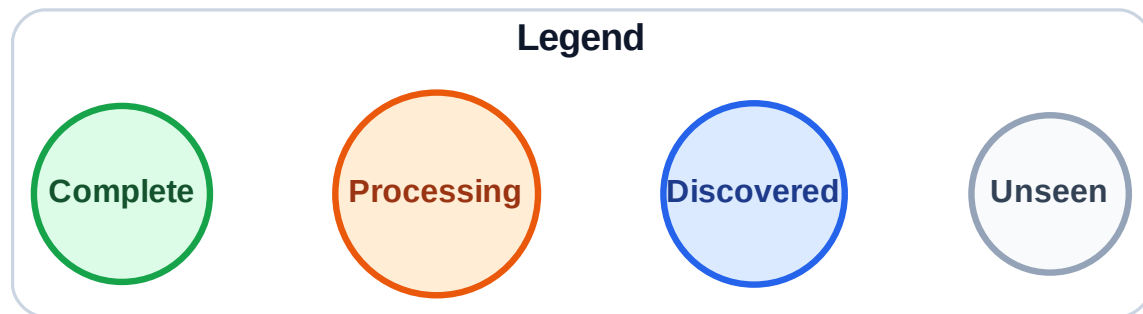


Queue: C → E → F → M → K
Visited: A, B

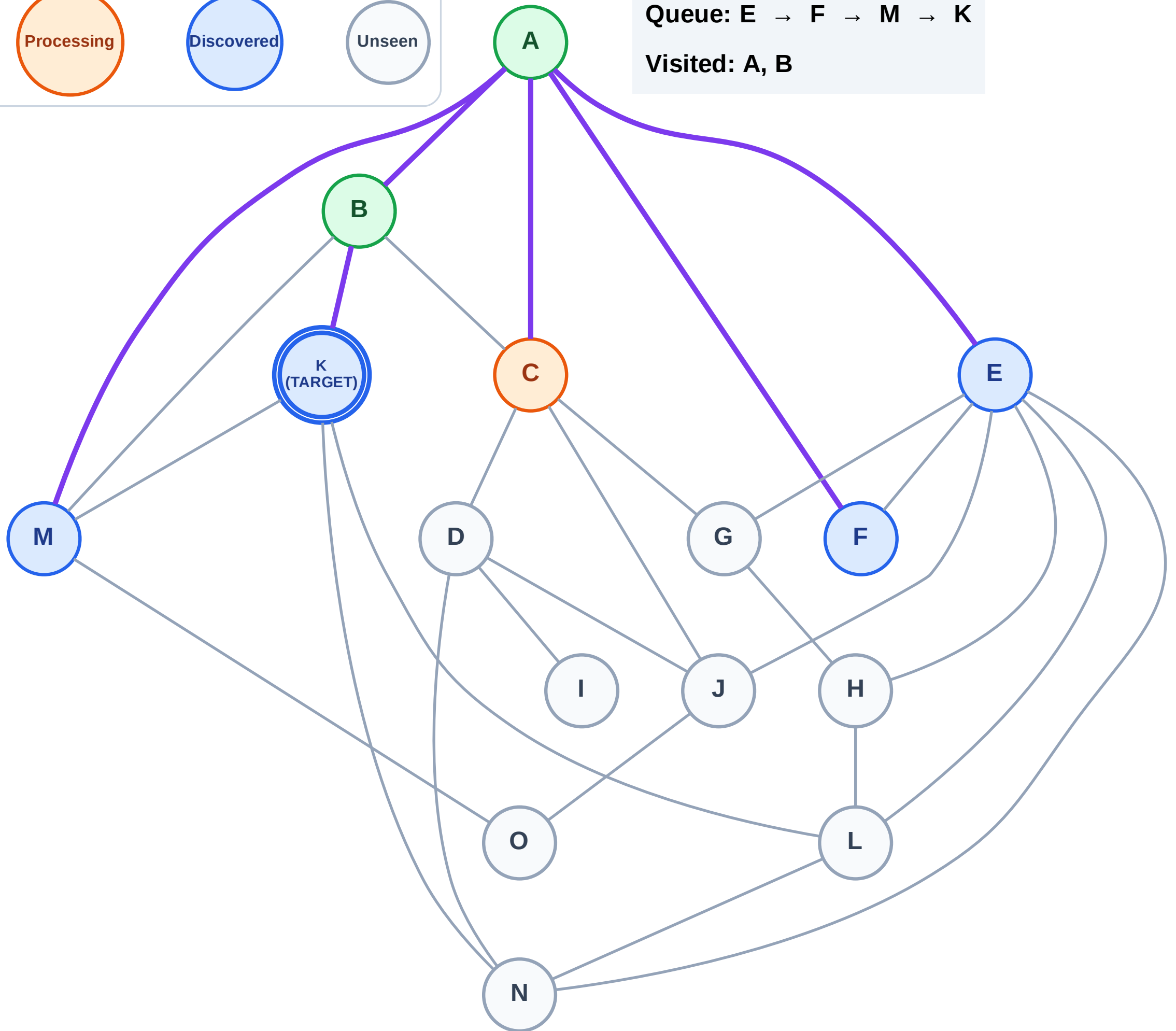


BFS Traversal

Step 6: Process node C



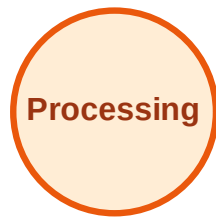
Queue: E → F → M → K
Visited: A, B



BFS Traversal

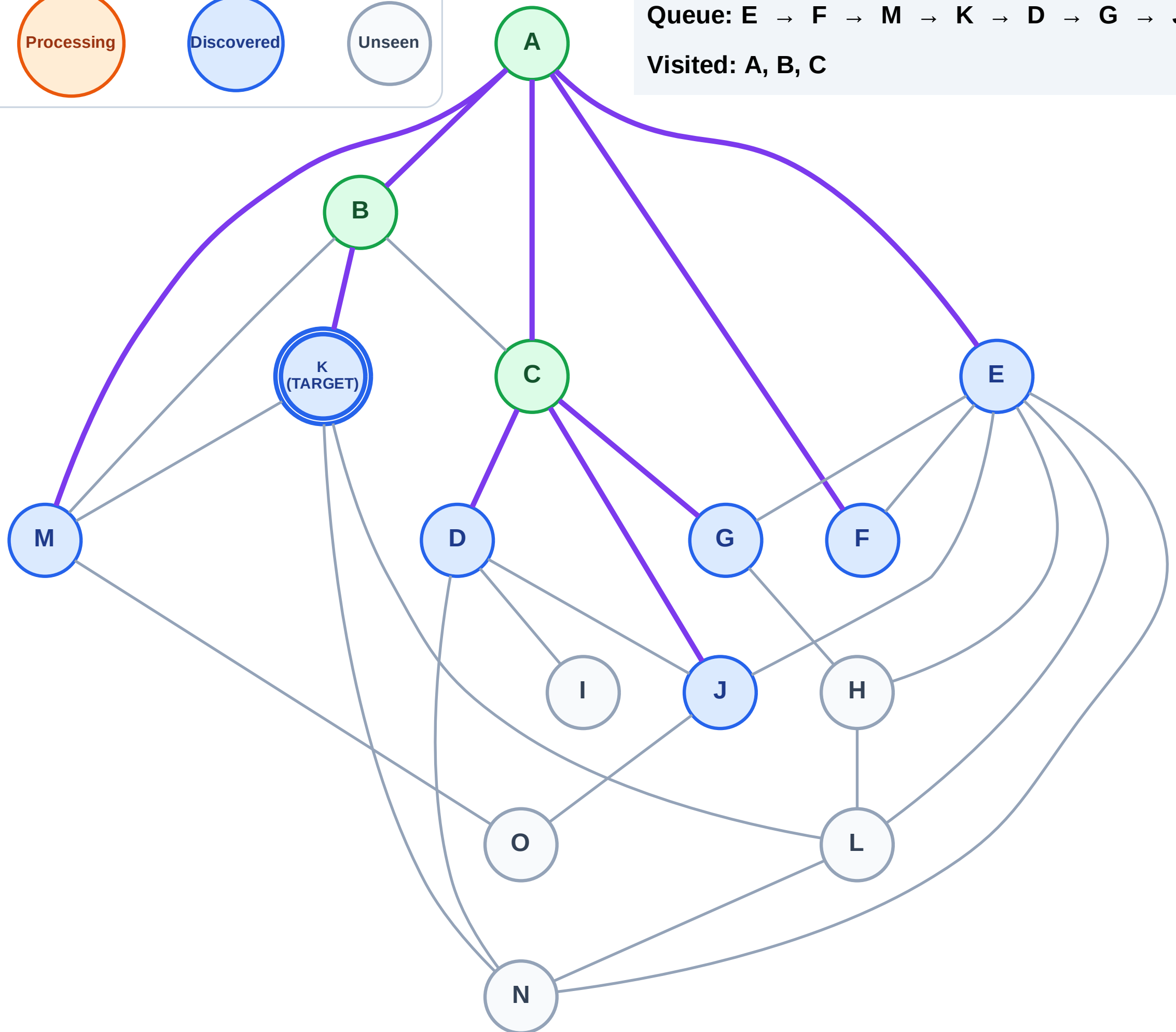
Step 7: Discover D, G, J from C

Legend

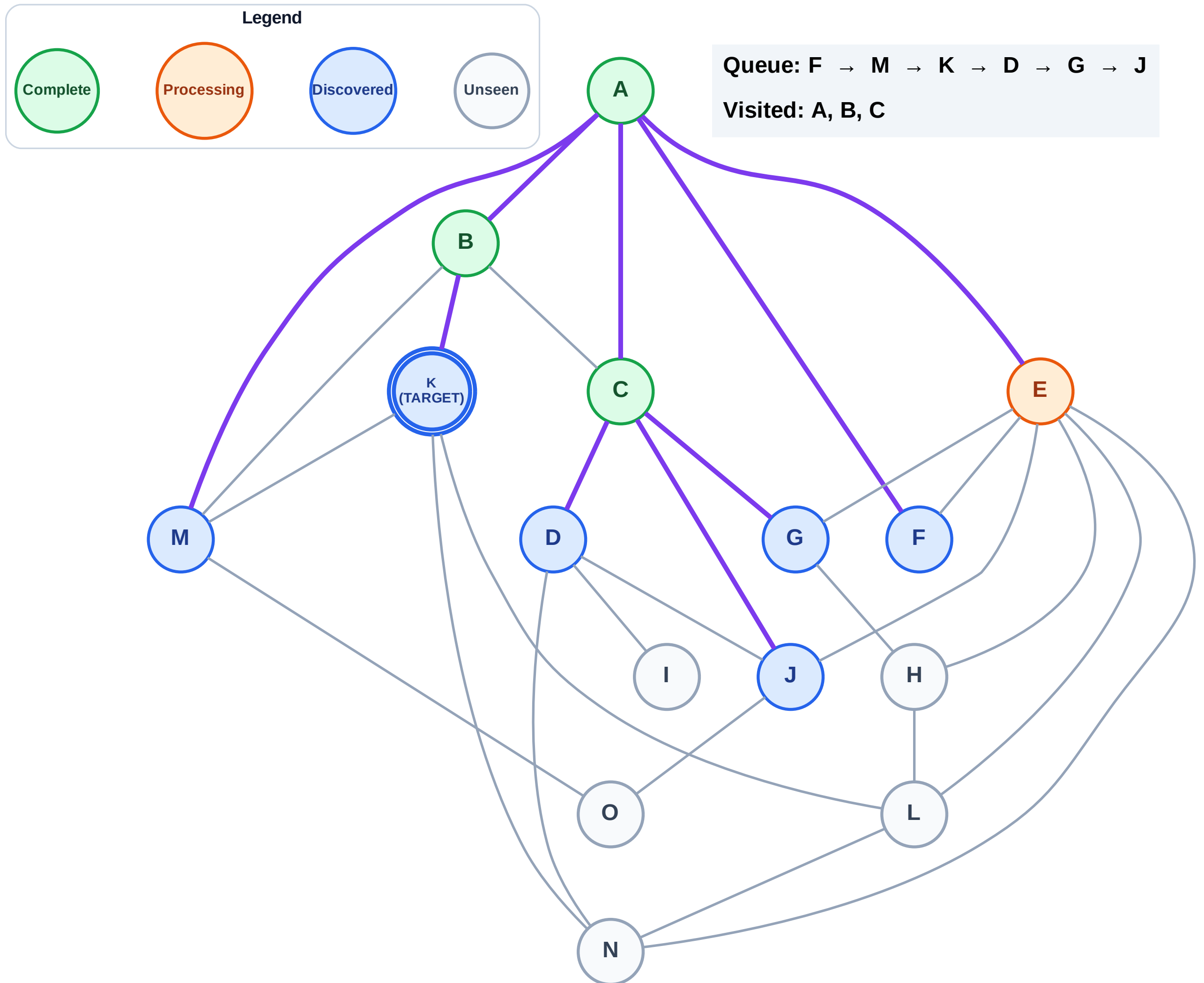


Queue: E → F → M → K → D → G → J

Visited: A, B, C



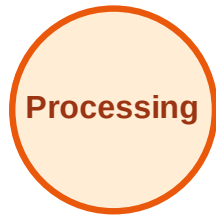
Step 8: Process node E



BFS Traversal

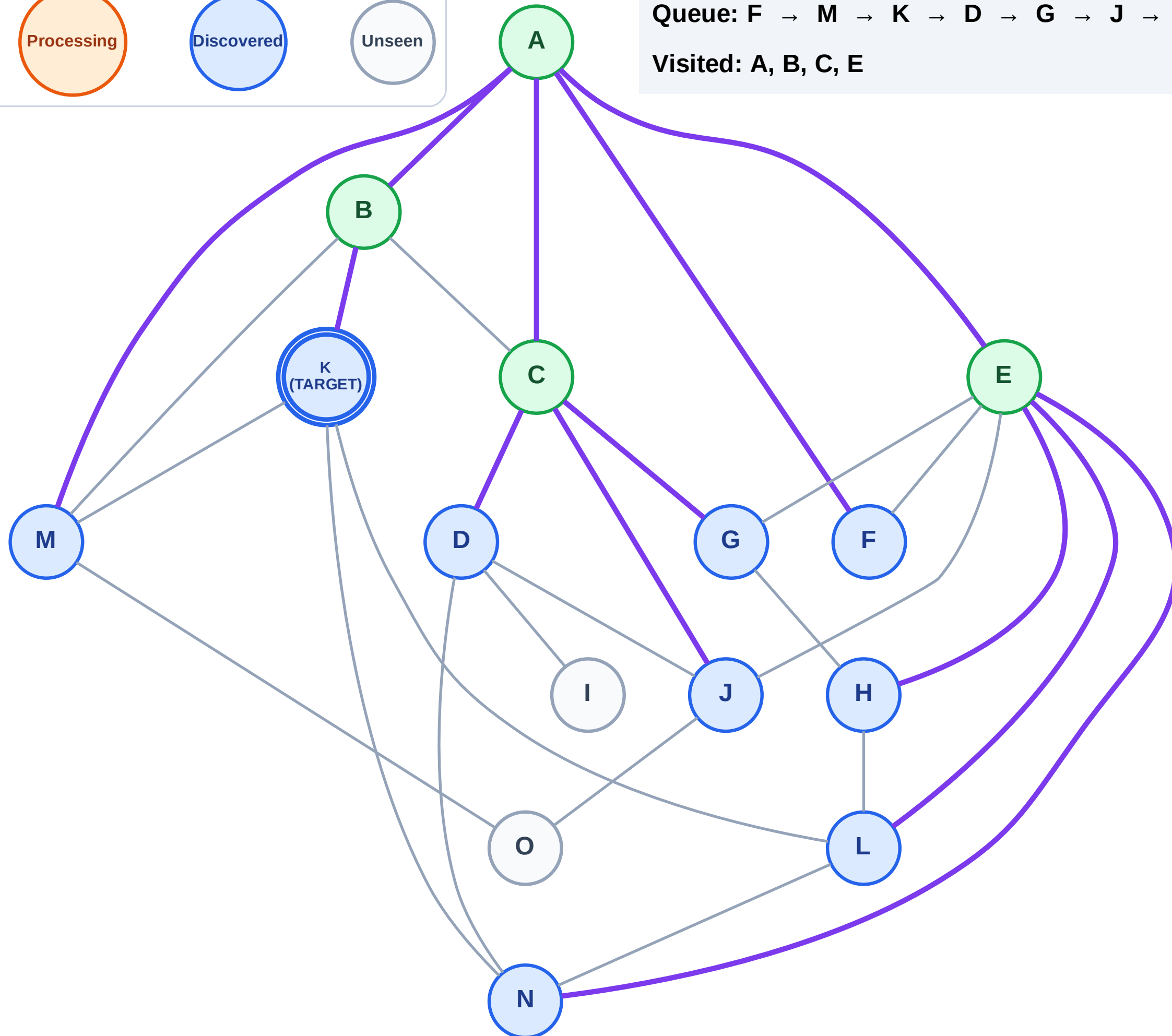
Step 9: Discover H, L, N from E

Legend



Queue: F → M → K → D → G → J → H → L → N

Visited: A, B, C, E



BFS Traversal

Step 10: Process node F

Legend

Complete

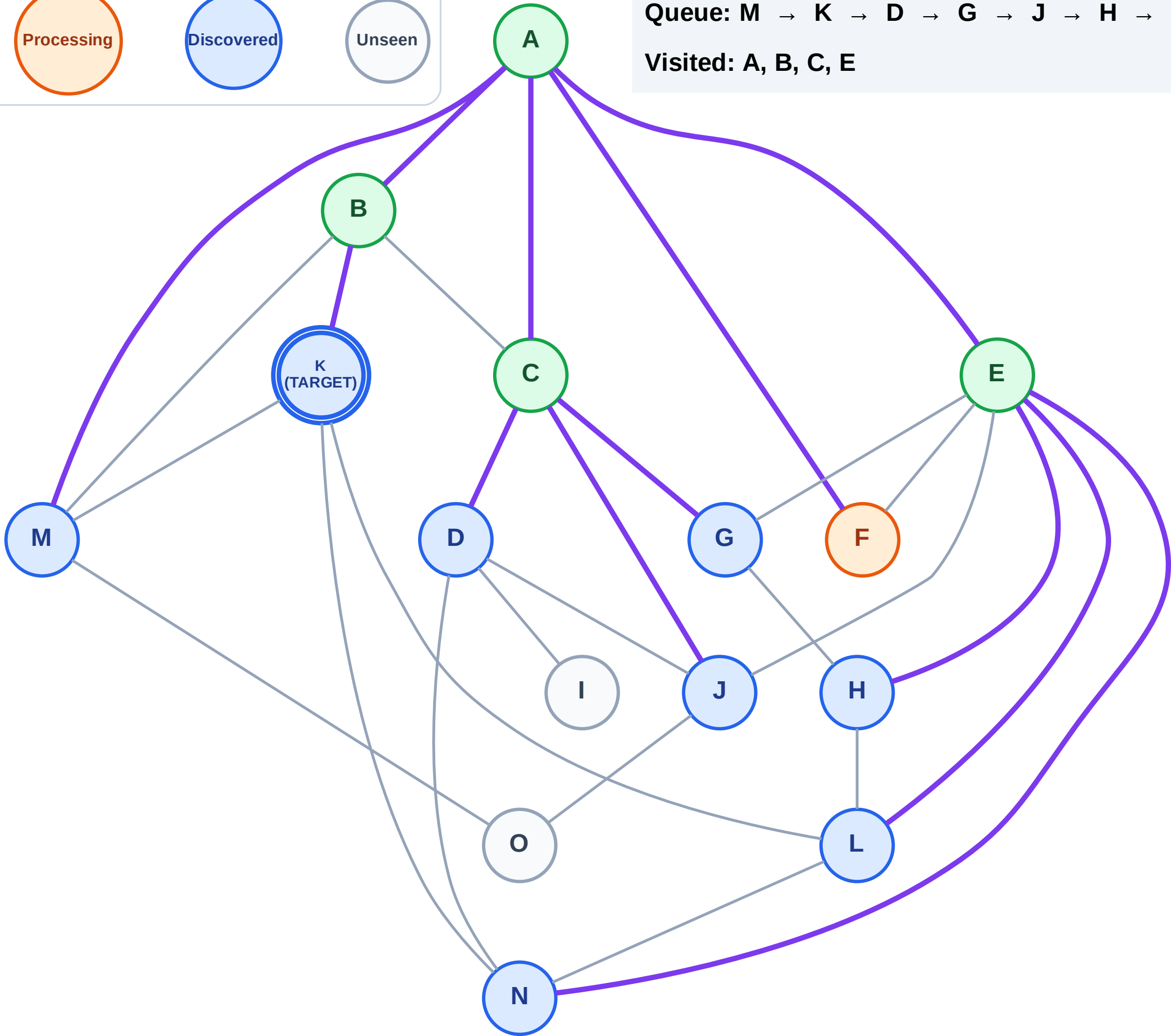
Processing

Discovered

Unseen

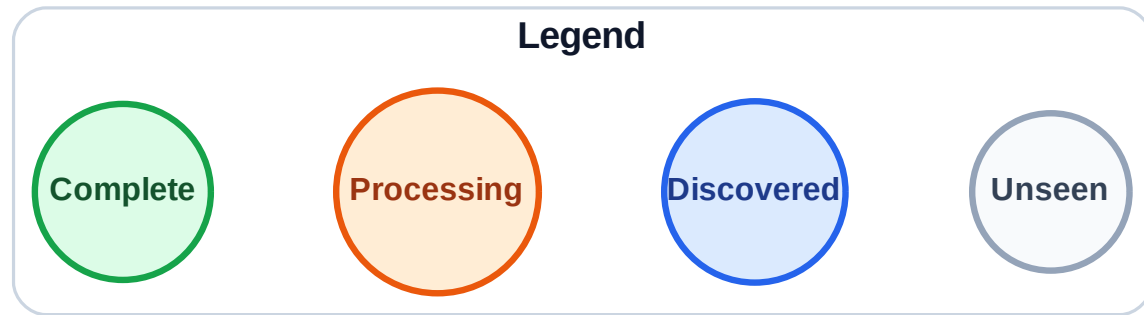
Queue: M → K → D → G → J → H → L → N

Visited: A, B, C, E



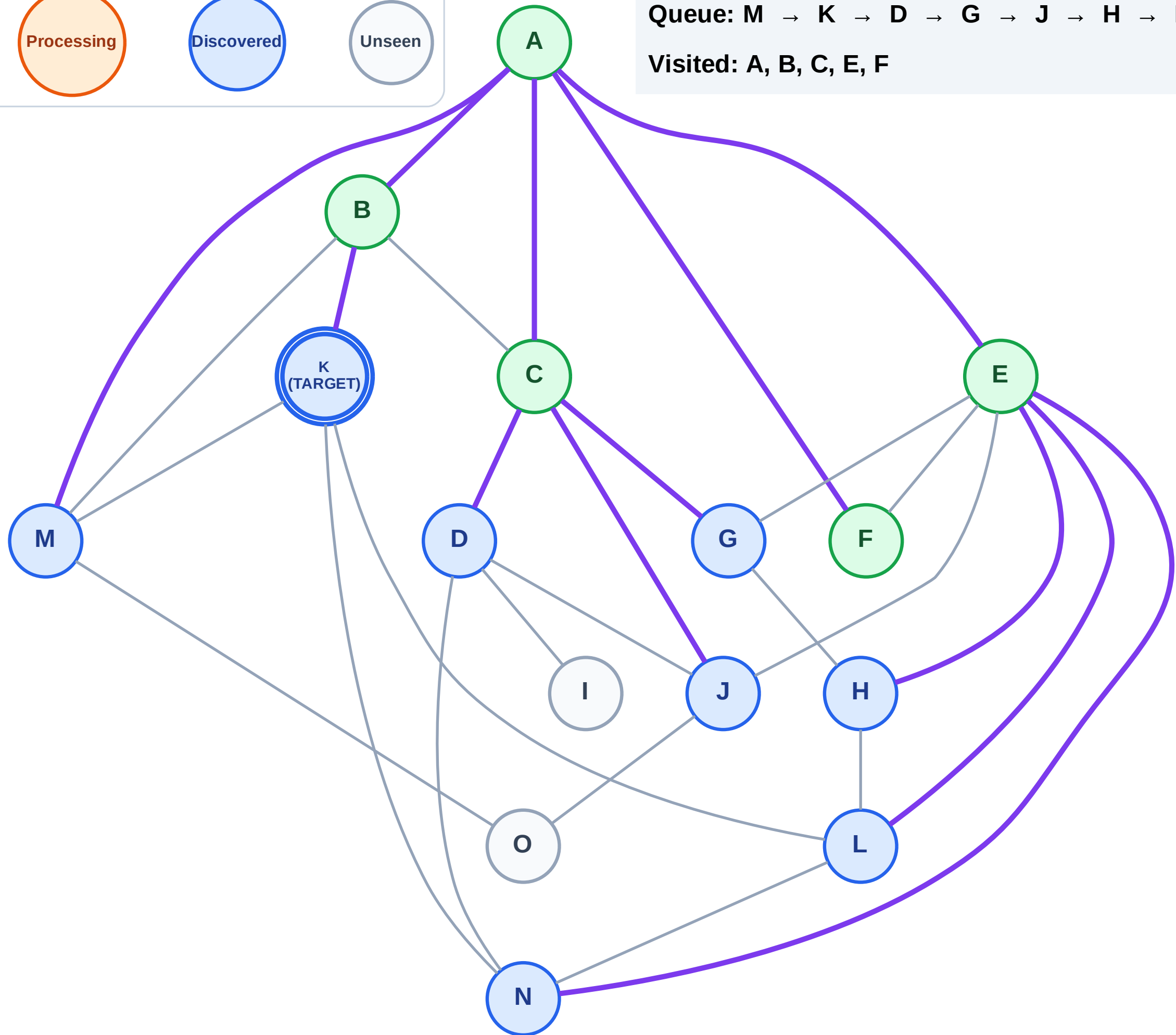
BFS Traversal

Step 11: No new neighbors from F



Queue: M → K → D → G → J → H → L → N

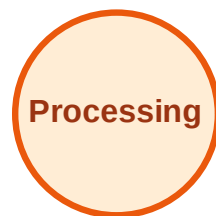
Visited: A, B, C, E, F



BFS Traversal

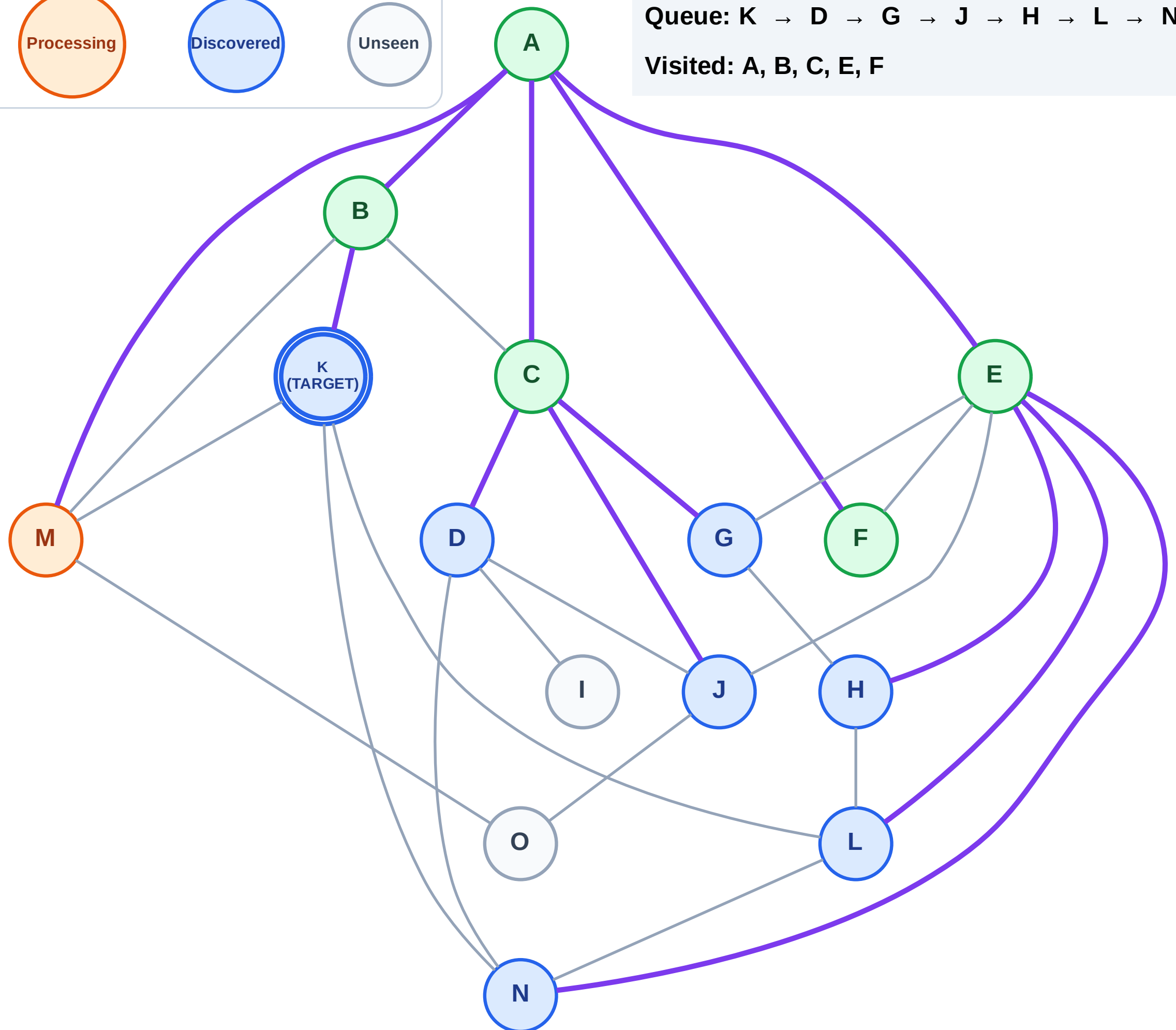
Step 12: Process node M

Legend



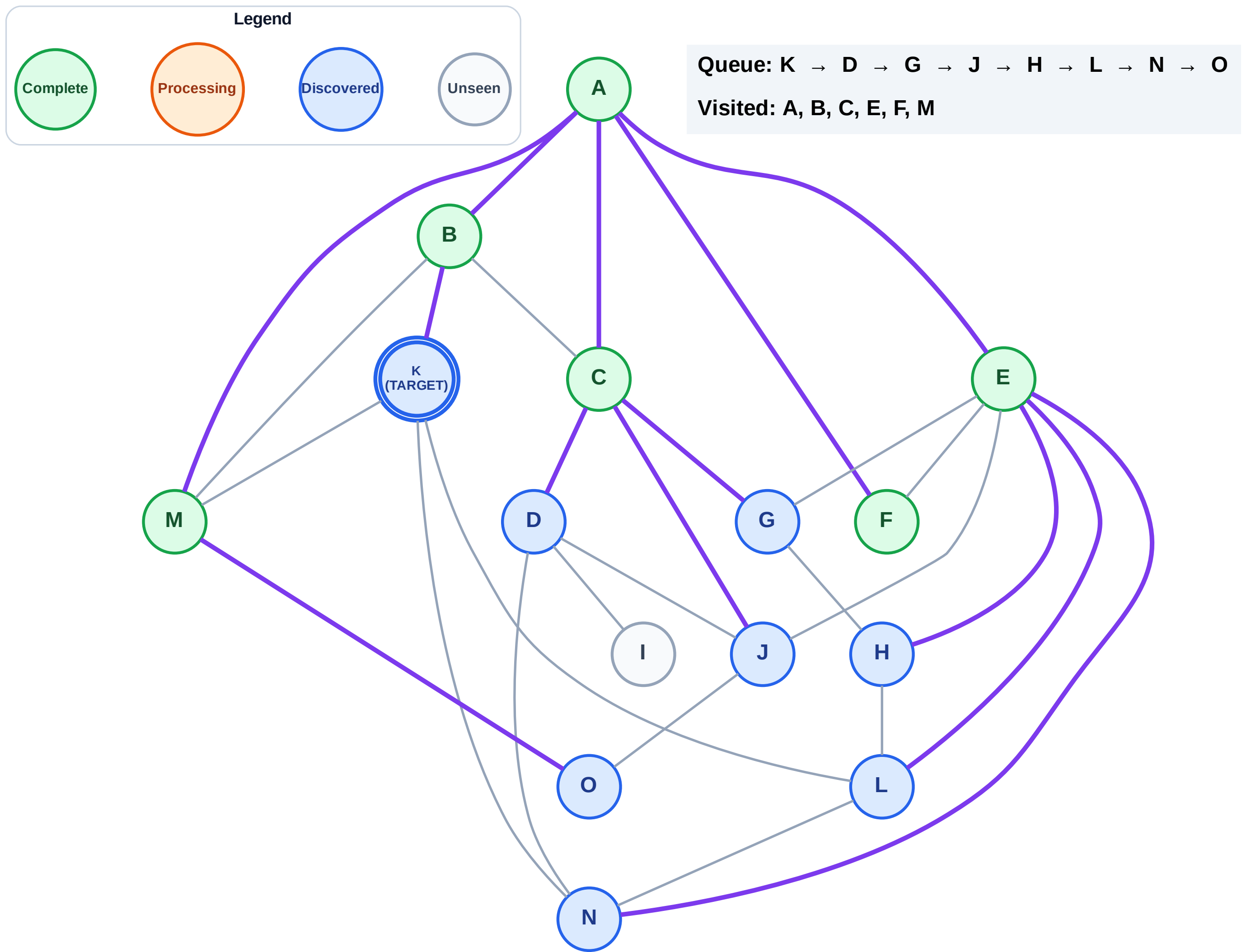
Queue: K → D → G → J → H → L → N

Visited: A, B, C, E, F



Step 13: Discover O from M

Step 13: Discover O from M

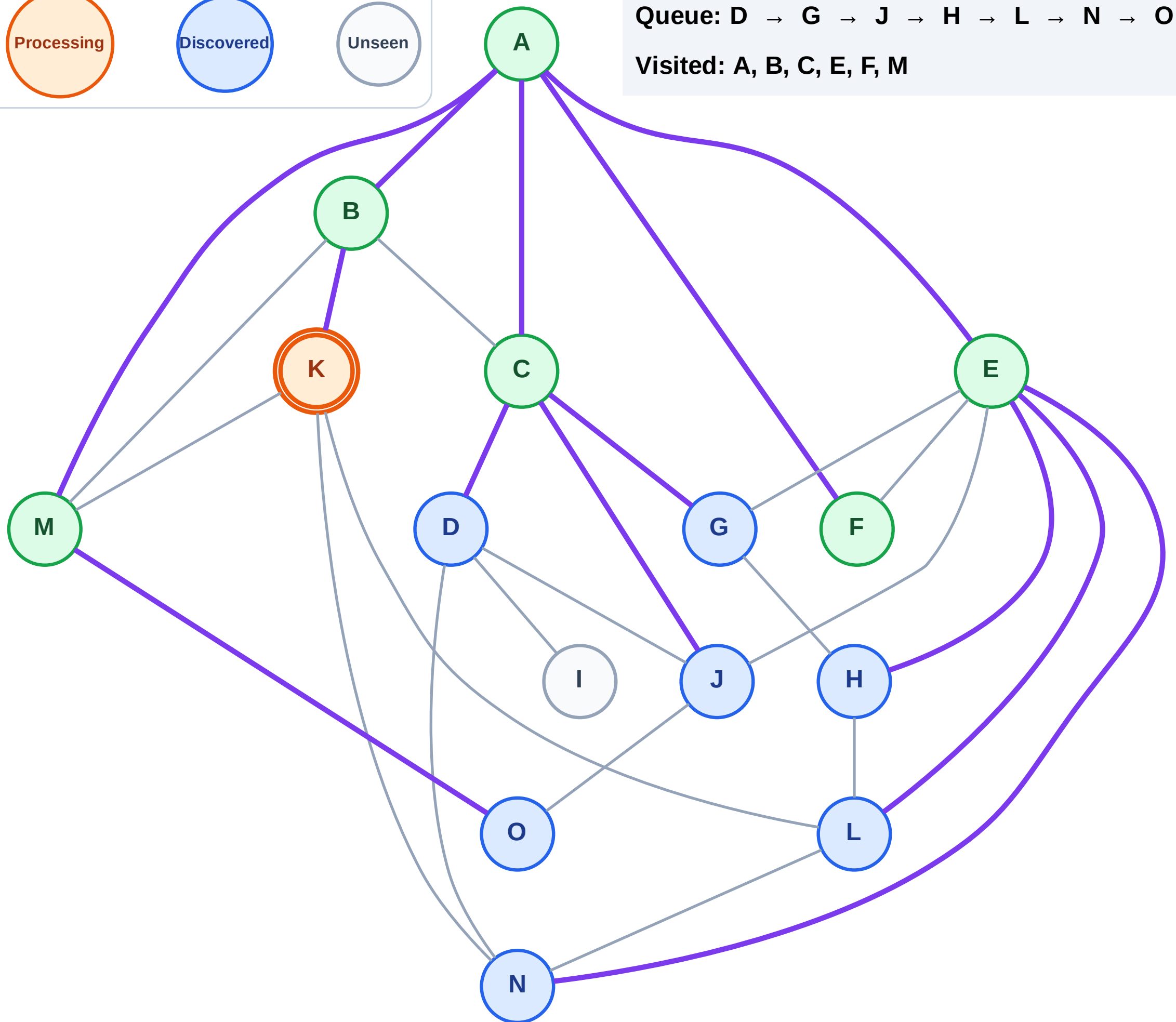


Step 14: Process node K

Legend

Unseen

Visited: A, B, C, E, F, M



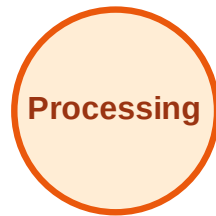
BFS Traversal

Step 15: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend



Complete



Processing



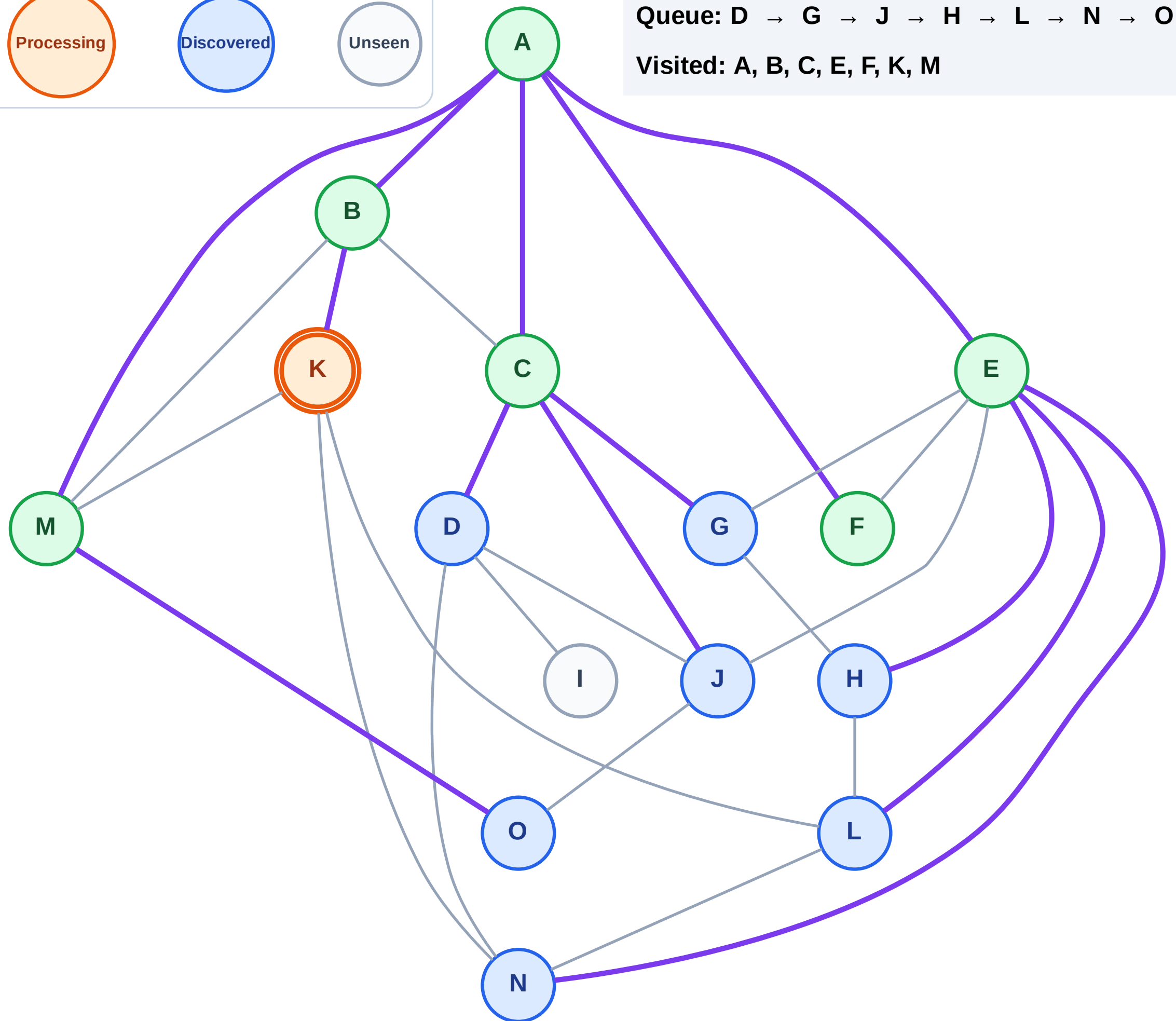
Discovered



Unseen

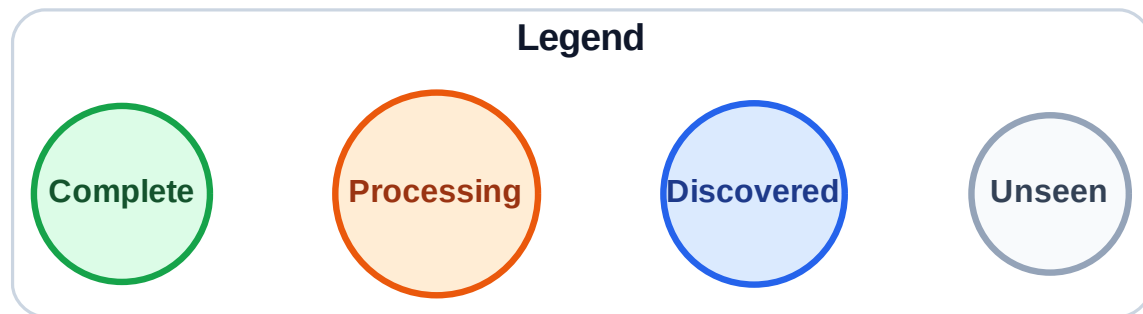
Queue: D → G → J → H → L → N → O

Visited: A, B, C, E, F, K, M

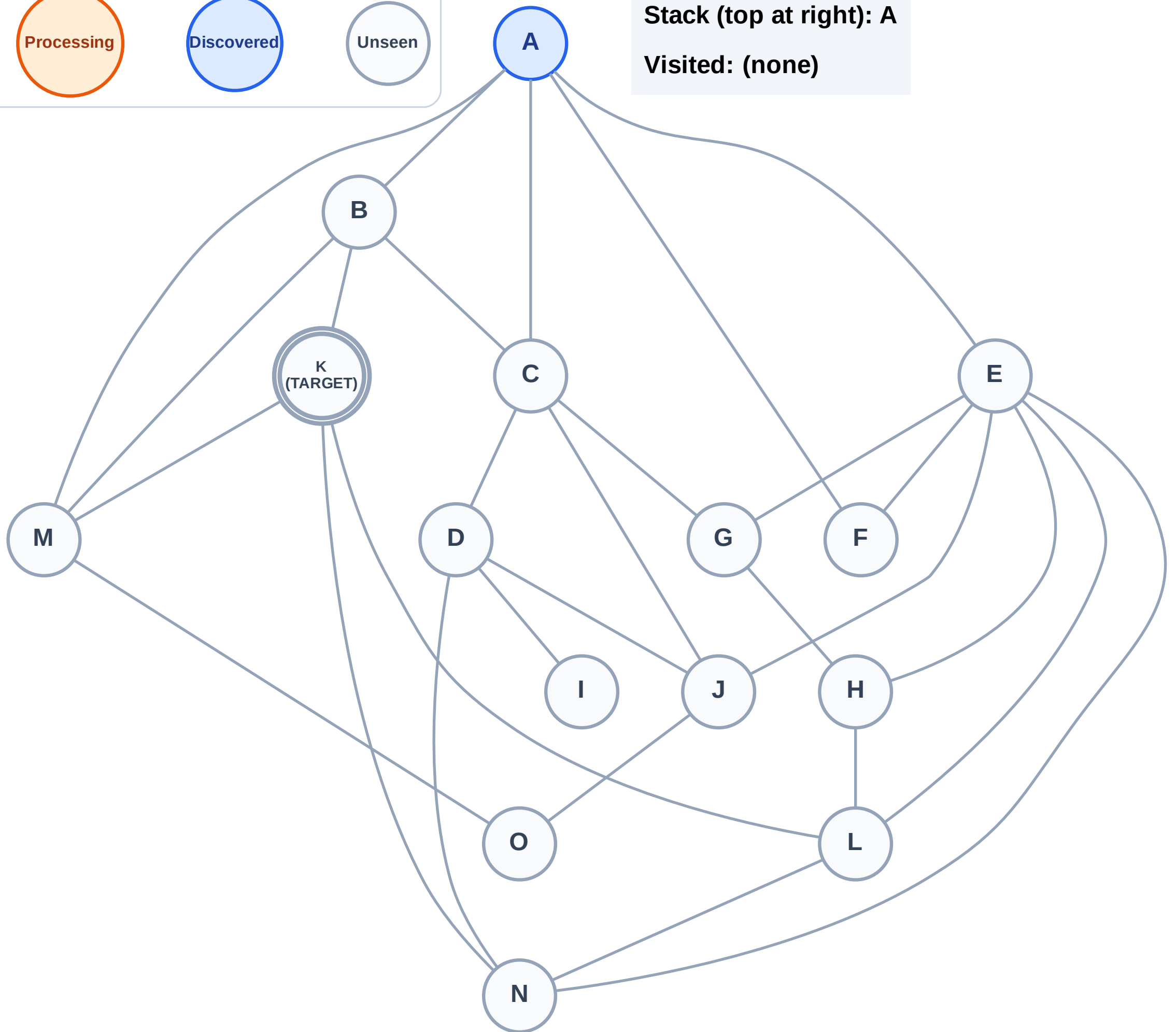


DFS Traversal

Step 1: Start at A (Seeking target node K)



Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

Complete

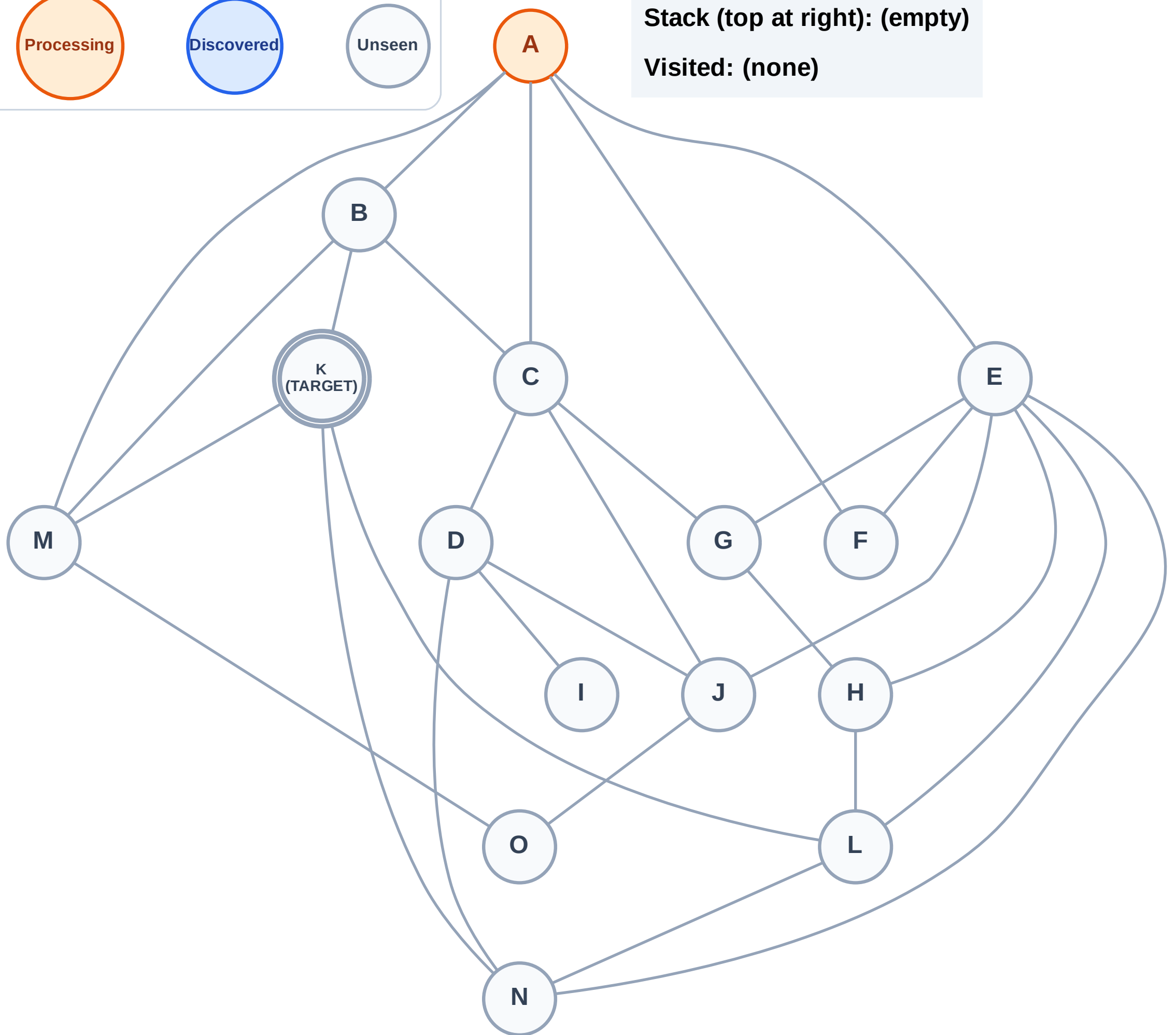
Processing

Discovered

Unseen

Stack (top at right): (empty)

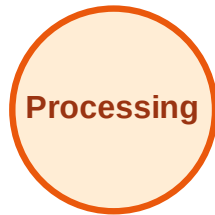
Visited: (none)



DFS Traversal

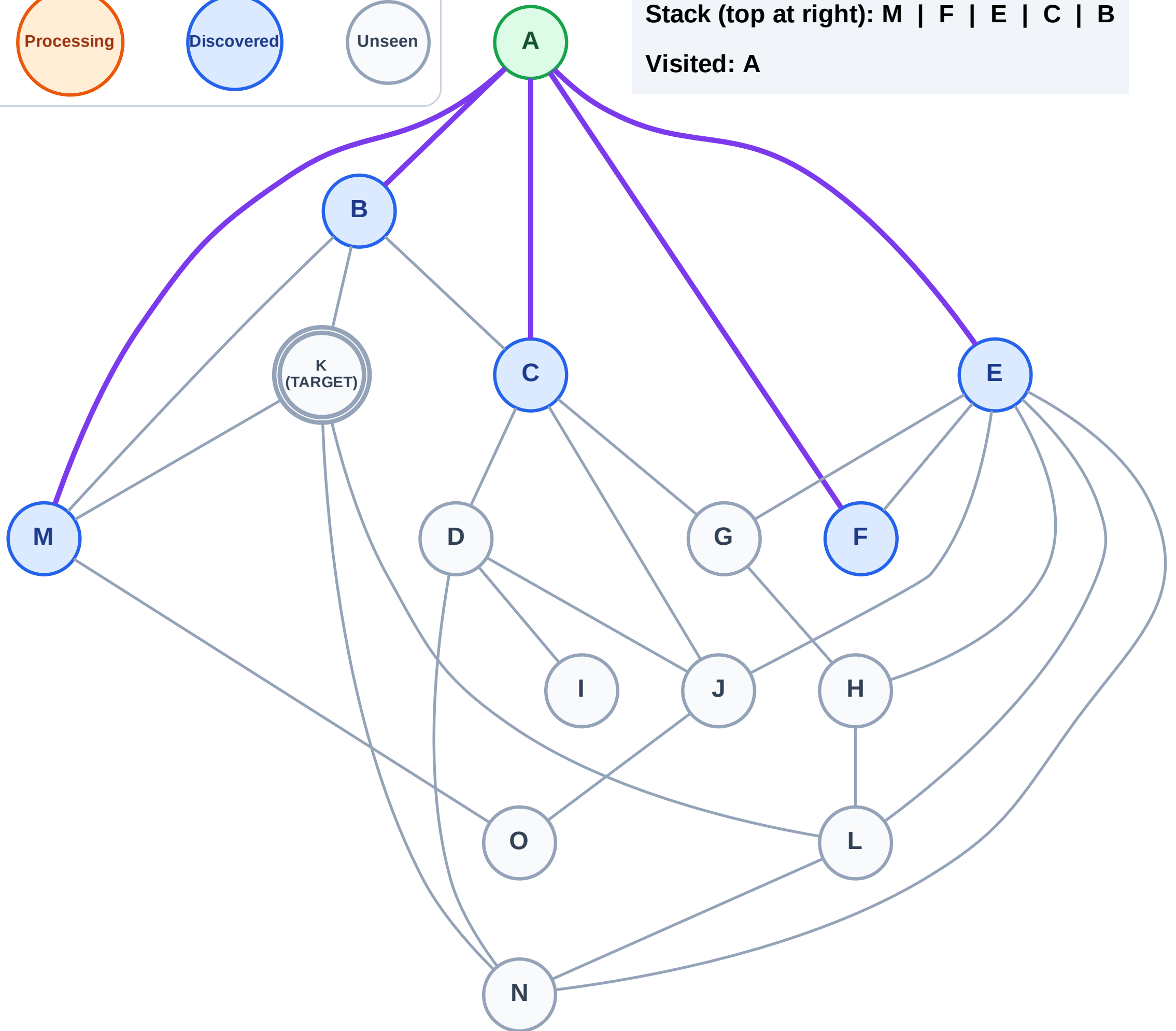
Step 3: Discover M, F, E, C, B from A

Legend



Stack (top at right): M | F | E | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

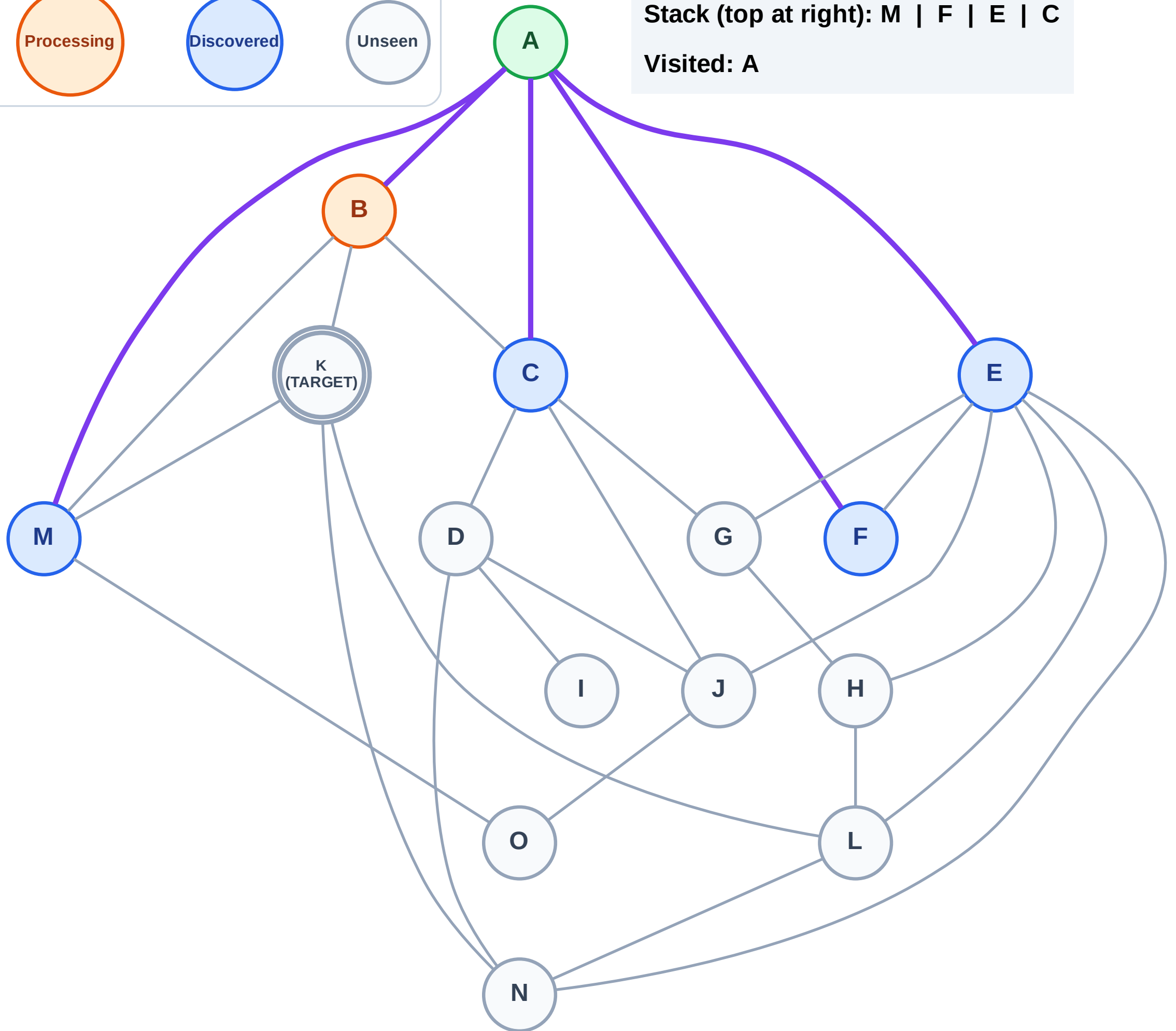
Processing

Discovered

Unseen

Stack (top at right): M | F | E | C

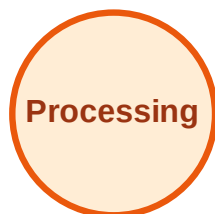
Visited: A



DFS Traversal

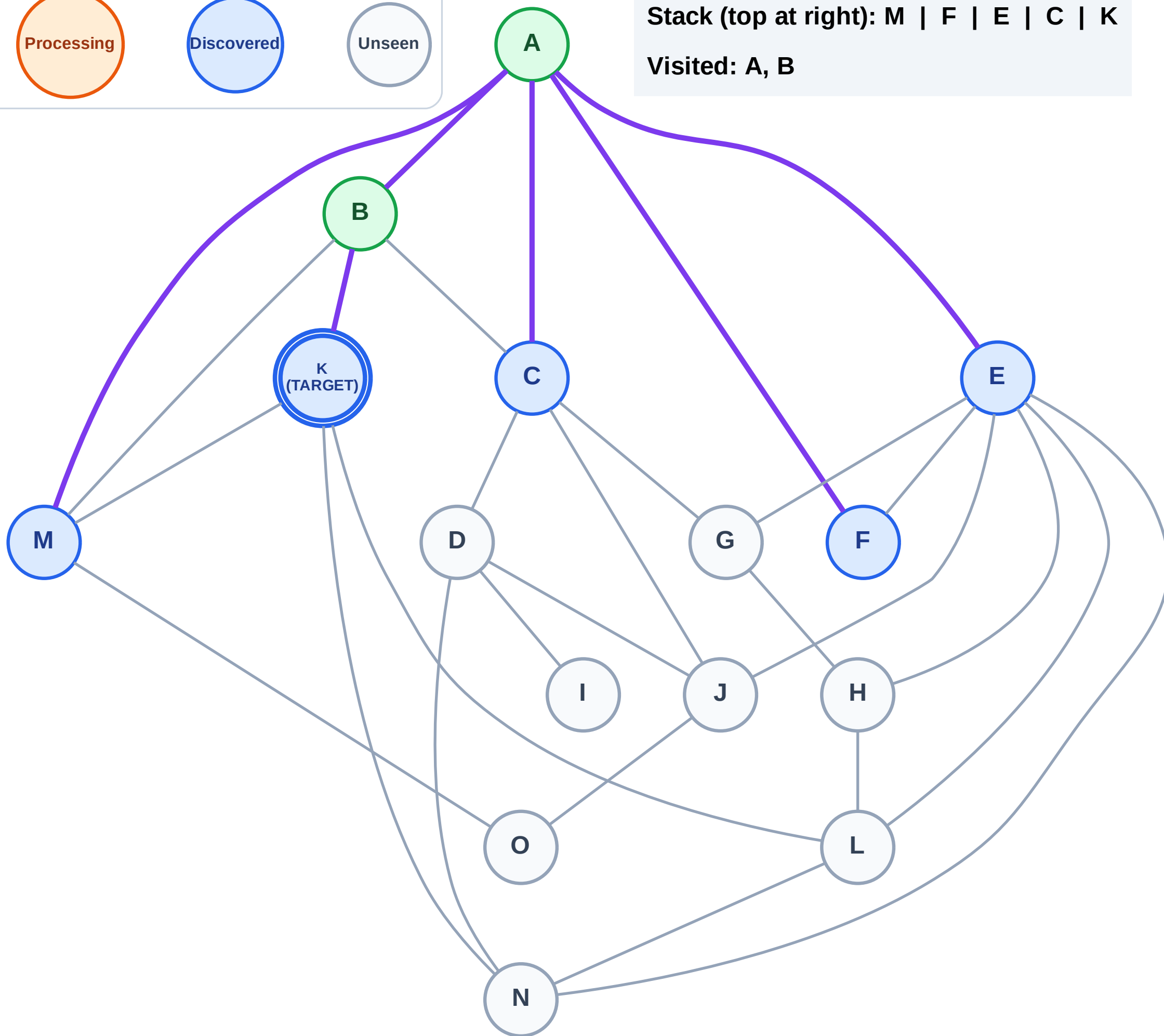
Step 5: Discover K from B

Legend



Stack (top at right): M | F | E | C | K

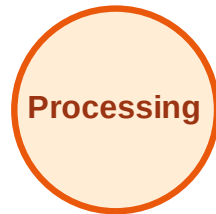
Visited: A, B



DFS Traversal

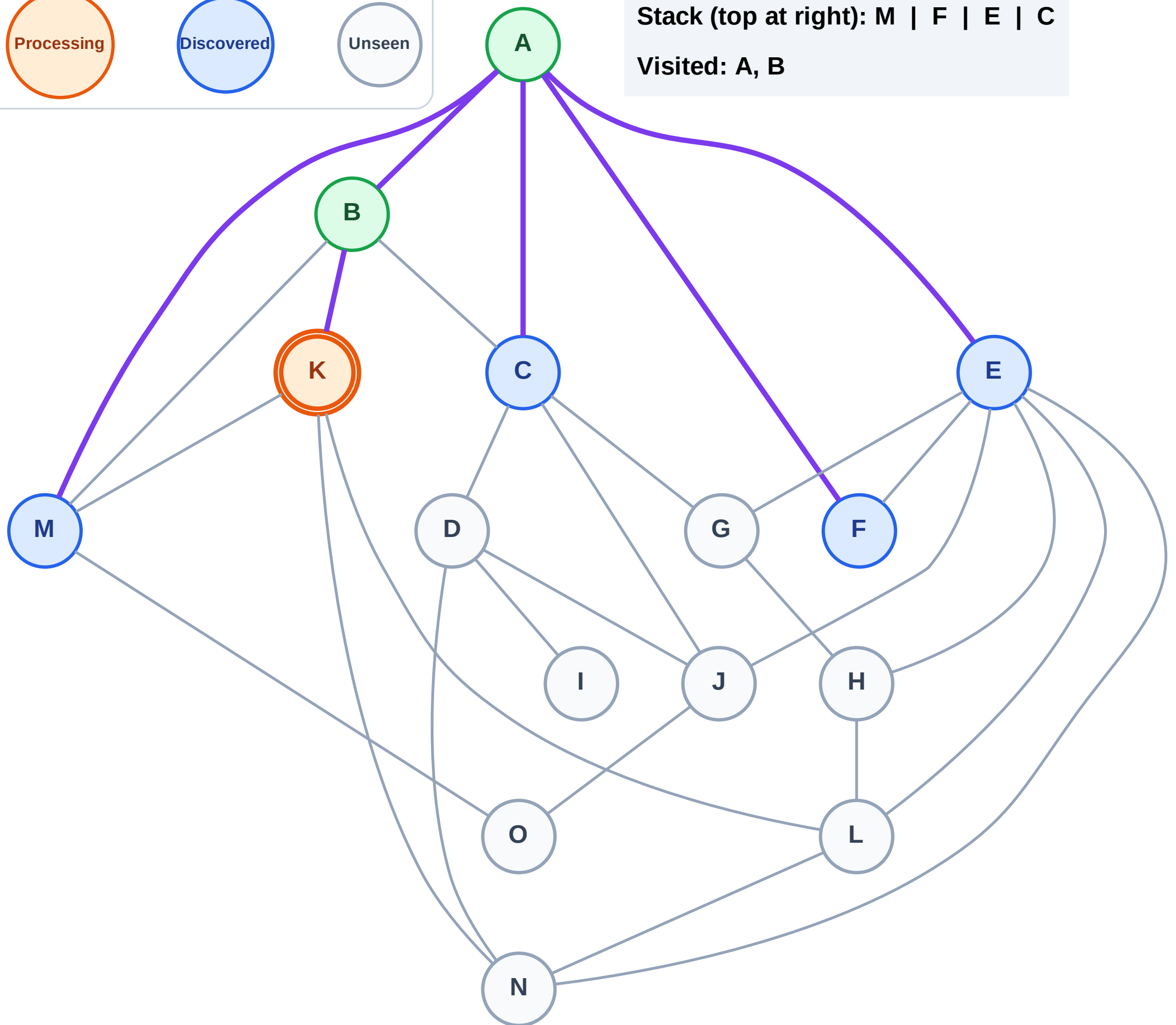
Step 6: Process node K

Legend



Stack (top at right): M | F | E | C

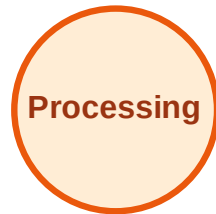
Visited: A, B



DFS Traversal

Step 7: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend



Stack (top at right): M | F | E | C

Visited: A, B, K

